CHAPTER EXERCISE: Answers for this chapter start on page 284.

A calculator should NOT be used on the following questions.

1

Which of the following is equivalent to $6x^2y + 6xy^2$?

A) $6xy(x + y)$
B) $12xy(x + y)$
C) $6x^2y^2(y + x)$
D) $12x^3y^3$

2

If $a \neq 0$, then $\frac{1}{a} + \frac{3}{4}$ is equivalent to which of the following?

A) $\frac{3 + 4a}{4a}$
B) $\frac{4 + 3a}{4a}$
C) $\frac{7}{4a}$
D) $\frac{4}{a + 4}$

3

Which of the following is equivalent to $(x^2 + y)(y + z)$?

A) $x^2z + y^2 + yz$
B) $x^2y + x^2z + y^2 + yz$
C) $x^2y + y^2 + x^2z$
D) $x^2 + x^2z + y^2 + yz$

4

Which of the following is equivalent to $\frac{4 + 8x}{12x}$ for $x \neq 0$?

A) $\frac{1 + 8x}{3x}$
B) $\frac{4 + 2x}{3x}$
C) $\frac{1 + 2x}{3x}$
D) 1

5

Which of the following is equivalent to $3x^4 - 3$?

A) $3(x^2 + 1)^2$
B) $3(x^2 - 1)^2$
C) $3(x^3 - 1)(x + 1)$
D) $3(x^2 + 1)(x + 1)(x - 1)$

6

$$(x + 1)^2 + 2(x + 1)(y + 1) + (y + 1)^2$$

Which of the following is equivalent to the expression shown above?

A) $(x + y + 1)^2$
B) $(x + y + 2)^2$
C) $(x + y)^2 + 2$
D) $(x + y)^2 - x - y$

7

If $y \neq 0$ and $x \neq y$, which of the following is equivalent to $\dfrac{xy - x^2}{xy - y^2}$?

A) $-\dfrac{y}{x}$

B) $\dfrac{y}{x}$

C) $\dfrac{x}{y}$

D) $-\dfrac{x}{y}$

8

If $x > 1$, which of the following is equivalent to $\dfrac{1}{\dfrac{x-1}{2} + \dfrac{x+5}{3}}$?

A) $\dfrac{5x+7}{6}$

B) $\dfrac{6}{2x+4}$

C) $\dfrac{6}{5x+7}$

D) $\dfrac{1}{30x+42}$

9

$$\frac{2 + \dfrac{1}{x}}{2 - \dfrac{1}{x}}$$

If $x \neq 0$, which of the following is equivalent to the given expression?

A) $\dfrac{2x-1}{2x+1}$

B) $\dfrac{2x+1}{2x-1}$

C) $\dfrac{4x^2-1}{x^2}$

D) -1

10

The expression $8x^2 - \dfrac{1}{2}y^2$ can be written in the form $8(x - cy)(x + cy)$, where c is a positive constant. What is the value of c ?

A) $\dfrac{1}{16}$

B) $\dfrac{1}{8}$

C) $\dfrac{1}{4}$

D) $\dfrac{\sqrt{2}}{4}$

11

$$x^2(x+2)(x-2) + 4$$

Which of the following is equivalent to the expression above?

A) $(x^2-2)^2$

B) $(x^2+2)^2$

C) $(x-1)^2(x+2)^2$

D) $(x+1)^2(x-2)^2$

A calculator is allowed on the following questions.

12

$$3x^3 + 8x^2 - 4x$$

$$7x^2 - 11x - 7$$

Which of the following is the sum of the two polynomials above?

A) $3x^3 + x^2 - 15x - 7$
B) $3x^3 + 15x^2 - 15x - 7$
C) $10x^5 - 7x - 7$
D) $15x^4 + 3x^3 - 15x^2 - 7$

13

$$(5a + 3\sqrt{a}) - (2a + 5\sqrt{a})$$

Which of the following is equivalent to the expression above?

A) $-2a\sqrt{a}$
B) $a\sqrt{a}$
C) $3a - 2\sqrt{a}$
D) $3a + 8\sqrt{a}$

14

If $y \neq 0$, what is the value of $\dfrac{9(2y)^2 + 2(6y)^2}{8(3y)^2}$?

15

$$\frac{x}{x-2} + \frac{x}{2(2-x)}$$

Which of the following is equivalent to the expression above for $x \neq 2$?

A) $-\dfrac{x}{x-2}$
B) $-\dfrac{x}{2(x-2)}$
C) $\dfrac{x}{2(x-2)}$
D) $\dfrac{3x}{2(x-2)}$

Chapter 6: Expressions

CHAPTER EXERCISE:

1. [A] We factor out $6xy$ from both terms to get $6xy(x+y)$.
2. [B] The least common denominator is $4a$. So, $\frac{1}{a}+\frac{3}{4}=\frac{4}{4a}+\frac{3a}{4a}=\frac{4+3a}{4a}$
3. [B] Expanding, $(x^2+y)(y+z)=x^2y+x^2z+y^2+yz$
4. [C] Divide the top and bottom by 4 to get $\frac{1+2x}{3x}$. Another way to get the same answer is to split the fractions and reduce.
5. [D] $3x^4-3=3(x^4-1)=3(x^2+1)(x^2-1)=3(x^2+1)(x+1)(x-1)$
6. [B] The expression follows the $(a+b)^2=a^2+2ab+b^2$ pattern, where $a=x+1$ and $b=y+1$. Therefore, the expression is equivalent to $((x+1)+(y+1))^2=(x+y+2)^2$.
7. [D] $\frac{xy-x^2}{xy-y^2}=\frac{x(y-x)}{y(x-y)}=\frac{-x(x-y)}{y(x-y)}=-\frac{x}{y}$
8. [C] Adding the two fractions in the denominator, $\frac{x-1}{2}+\frac{x+5}{3}=\frac{3(x-1)+2(x+5)}{6}=\frac{5x+7}{6}$. Now, 1 over this result means we can flip it: $\frac{6}{5x+7}$.
9. [B] $\frac{2+\frac{1}{x}}{2-\frac{1}{x}}=\frac{\frac{2x}{x}+\frac{1}{x}}{\frac{2x}{x}-\frac{1}{x}}=\frac{\frac{2x+1}{x}}{\frac{2x-1}{x}}=\frac{2x+1}{x}\times\frac{x}{2x-1}=\frac{2x+1}{2x-1}$
10. [C] First factor out an 8 from both terms. Then use the formula $a^2-b^2=(a-b)(a+b)$.

$$8x^2-\frac{1}{2}y^2=8\left(x^2-\frac{1}{16}y^2\right)=8\left(x-\frac{1}{4}y\right)\left(x+\frac{1}{4}y\right)$$

Therefore, $c=\frac{1}{4}$.

11. [A] First, expand: $x^2(x+2)(x-2)+4=x^2(x^2-4)+4=x^4-4x^2+4$. Now we can apply the formula $a^2-2ab+b^2=(a-b)^2$, where $a=x^2$ and $b=2$, to get $x^4-4x^2+4=(x^2-2)^2$.
12. [B] Combining like terms, we get $3x^3+(8x^2+7x^2)+(-4x-11x)-7=3x^3+15x^2-15x-7$.
13. [C] Combining like terms, $5a-2a=3a$ and $3\sqrt{a}-5\sqrt{a}=-2\sqrt{a}$.
14. [$\frac{3}{2}$] $\frac{9(2y)^2+2(6y)^2}{8(3y)^2}=\frac{36y^2+72y^2}{72y^2}=\frac{1}{2}+1=\frac{3}{2}$
15. [C] To get a common denominator of $2(x-2)$ for both fractions, we first factor out a negative from the second fraction and then we multiply the top and bottom of the first fraction by 2:

$$\frac{x}{x-2}+\frac{x}{2(2-x)}=\frac{x}{x-2}+\frac{x}{-2(x-2)}=\frac{x}{x-2}-\frac{x}{2(x-2)}=\frac{2x}{2(x-2)}-\frac{x}{2(x-2)}=\frac{x}{2(x-2)}$$